Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_direct_generate_plus_one_repair.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	43.36386	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	76.84996	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1421.75094	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	1.0	0.073643	0.073643	195.1683	2	0	0	0.761107	12.115
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	198.23672	2	0	0	0.961836	14.385
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.273435	0.273435	142.62986	0	0	0	0.0	0.0

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 43.36386.
- Accepted epoch count: 0.
- Fallback epoch count: 0.

- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 76.84996.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1421.75094.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:

- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.069472.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073643.
- Held-out runtime ms: 195.1683.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002874.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.099117, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.096475, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.067432, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.047481, errors=[]

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 198.23672.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
- X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 142.62986.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (conservative)

Aspect	Details
Conditions Evaluated (6 total)	Baselines: Nearest Neighbor, Clarke-Wright Savings, Regret Insertion Local Search; Learned: Direct Generate + One Repair, Budget-matched No Replay, Replay Solver Evolution
Direct Synthesis	*Direct Generate + One Repair* achieved 100% feasibility with a heldout feasible gap of 0.0736, close to Clarke-Wright baseline (0.0738) but with higher runtime (~195 ms vs. 77 ms) and higher solution complexity/novelty
Budget-Matched Independent Search	*Budget-matched No Replay* failed to produce feasible solutions on heldout data (0% feasibility, penalized gap 3.0), despite longer runtime (~198 ms)
Replay-Aware Iterative Search	*Replay Solver Evolution* maintained 100% feasibility but with higher feasible gap (0.2734), similar to Nearest Neighbor baseline, showing no improvement
Fixed-Baseline Context	Clarke-Wright Savings baseline provided best tradeoff among baselines: 100% feasibility, 0.0738 gap, 77 ms runtime; Regret Insertion had slower runtime (1422 ms) and worse gap (0.4664)
Feasibility	All conditions except Budget-Matched No Replay maintained perfect (100%) feasibility on heldout data

Aspect	Details
Objective Gap	Best gap: Direct Generate + One Repair (0.0736) marginally better than Clarke-Wright (0.0738), both far better than Nearest Neighbor and Regret Insertion
Runtime	Clarke-Wright fastest among good solutions (~77 ms), Direct Generate slower (~195 ms), Regret Insertion slowest (~1422 ms)
Robustness	Low dispersion (~0.0027-0.0078) for best solutions indicating consistent performance; Budget-Matched No Replay had zero feasibility indicating poor robustness
Learned Condition vs Fixed Baselines	*Direct Generate + One Repair* learned condition closely matched and slightly improved over Clarke-Wright baseline gap, with higher novelty/complexity but no clear dominance
Replay vs Budget-Matched No Replay	Replay (Solver Evolution) maintained feasibility but did not improve gap over baselines; budget-matched no replay failed feasibility—thus replay clearly beats no-replay budget-matched control in feasibility and robustness

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Summary Conclusion

- The learned *Direct Generate + One Repair* condition matches Clarke-Wright baseline feasibility and objective gap but at increased runtime and complexity; no learned condition clearly outperforms all fixed baselines.
- Replay-aware iterative search maintains feasibility and robustness, outperforming the budget-matched no-replay control which fails feasibility entirely.
- Baselines Clarke-Wright and Nearest Neighbor remain strong context points, with Regret Insertion dominated by longer runtime and worse gaps.